CATERPILLAR®

DEALER BEST PRACTICE SERIES

Bulk Fuel Cleanliness Management

Maintenance and Repair | Application | **Component Life Management** | Component Renewal (CRC) | MARC Management

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1.0 Introduction

The management of fuel cleanliness is essential for maximizing fuel injector life and reducing unplanned truck downtime. It is important to ensure proper contamination control throughout the entire fuel delivery process. This Best Practice will focus on maintaining fuel cleanliness from the moment it leaves the supplier to the time it is pumped into a machine's fuel tank.

Each stage in the process for delivering fuel to the equipment on a mine site needs to be evaluated in terms of ability to provide adequate contamination control. Things to assess at each stage include, but are not limited to; hoses, fast fill connectors, screen filters, caps, pumps, comptometers, level indicators, tanks, control valves, tank breathers and purge valves. Visually inspect and properly maintain these components, replacing when necessary. Frequently sample the fuel at each stage making sure its quality is ISO 18/16/13 or cleaner with a maximum water content of 1,000 ppm.

The following flowchart is an example of a process for fuel delivery to the equipment on site. As can be seen, there are many areas to ensure fuel quality and contamination control.

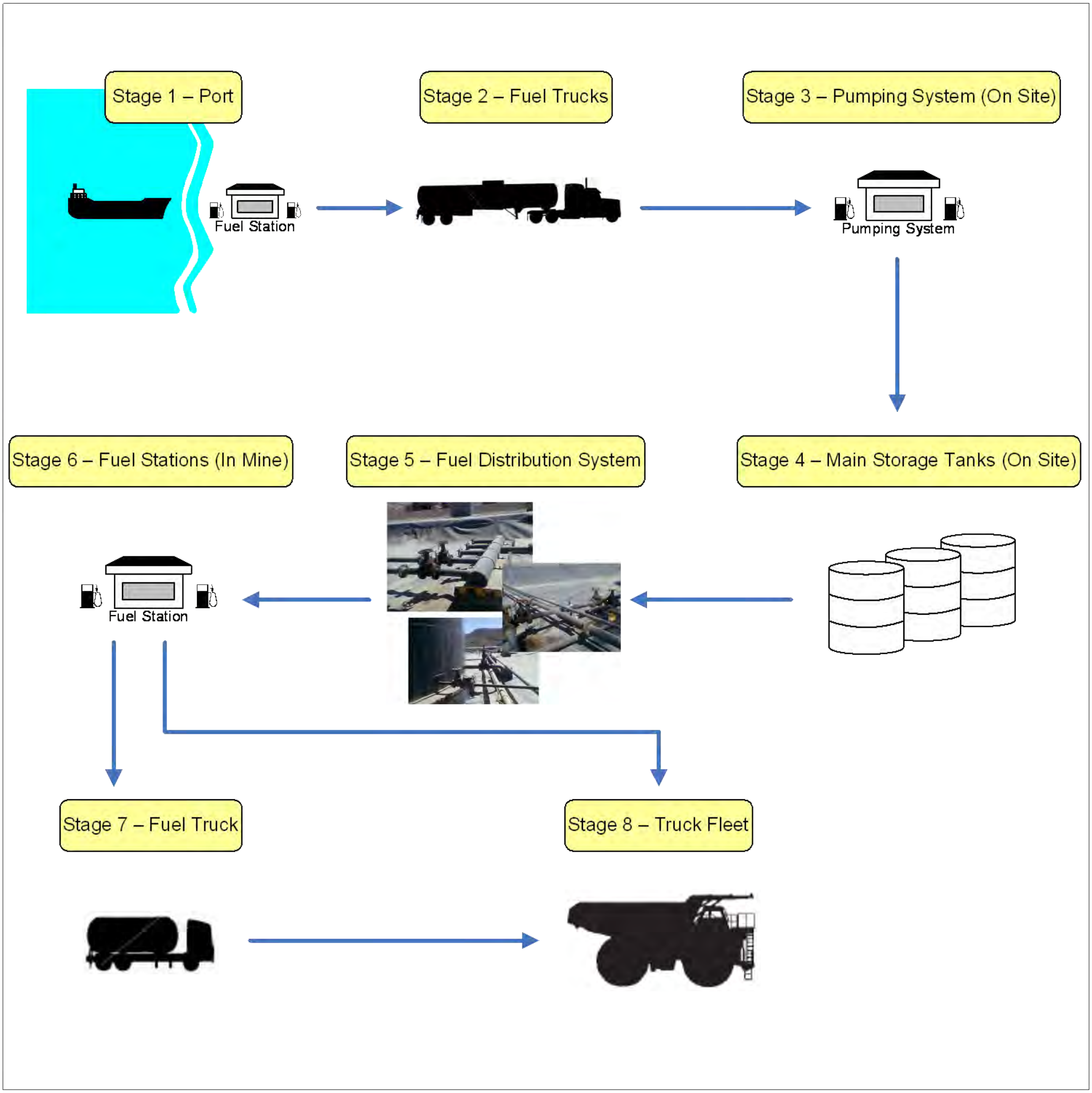

Figure 1 - Example Fuel Delivery Stages

2.0 Best Practice Description

Fuel cleanliness from suppliers is difficult to control from a contamination (debris & water) perspective. Fuel quality impacts injector wear/life resulting in engine power loss events and unplanned machine downtime. Contaminated fuel further impacts onboard filtration life. This scenario leads to excessive fuel consumption and often results in mid-life injector set replacement.

An improved onsite Bulk Fuel Filtration process will ensure fuel cleanliness, to reduce/eliminate power loss events, improve injector life, improve machine uptime, and further reduce filter change-out costs.

Clean fuel permits the machine’s onboard fuel filters to function properly without plugging. Injectors should also last to engine overhaul and provide improved long-term fuel economy/engine performance.

3.0 Implementation Steps

Assess the need to implement an on-site Bulk Fuel Filter Coalescer Unit based on supplier fuel quality and the presence of one on site. Technical information and pricing on Caterpillar Bulk Fuel Filter Coalescer Units can be obtained through Cat Global Mining and the Cat Filter and Fluids Group.

Analyze every stage in the fuel delivery process to the site to determine areas where the fuel can become contaminated. Identify personnel to inspect, maintain and repair/replace the contaminant barriers. Record fuel cleanliness at each stage to ensure ISO 18/16/13 standards are met and water content does not exceed 1,000 ppm. By recording the ISO levels both upstream and downstream of each stage will aid in determining if and where contaminants may be entering the system.

4.0 Benefits

Fuel cleanliness management has proven to provide the following:

- Reduce the number of injectors replaced per year.
- Reduce the number of on-board fuel filters and water separators replaced per year.
- Reduce the hours of unscheduled truck downtime.

This will result in cost savings on parts and labor and increase the utilization of the truck fleet.

A financial analysis of a real life scenario was performed to assign a dollar value to the benefits associated with maintaining fuel cleanliness. The results are summarized in the following table.

Fuel Filtration Financial Analysis

		Decreased O&O Costs		Increased Production		
		Year 1	*Year 2*	*Year 3*	*Year 4*	*Year 5*
789	*Fuel Filtration Program Net Benefit:*	$ 151,050	$ 151,050	$ 151,050	$ 151,050	$ 151,050
	O&O Cost Savings:	$ 14,603	$ 14,603	$ -	$ -	$ -
	Production Gains:	$ -	$ -	$ 161,498	$ 161,498	$ 161,498
793D	*Fuel Filtration Program Net Benefit:*	$ 103,959	$ 103,959	$ 103,959	$ 103,959	$ 103,959
	O&O Cost Savings:	$ 37,550	$ 37,550	$ -	$ -	$ -
	Production Gains:	$ -	$ -	$ 415,281	$ 415,281	$ 415,281
	Total:	**$ 307,162**	**$ 307,162**	**$ 831,789**	**$ 831,789**	**$ 831,789**

NPV @ 8%:	**$ 2,385,543**

Figure 2 – Example Benefits

In this example, the fleet consists of 789's and 793D's. The *'Fuel Filtration Program Net Benefits'* represents the cost savings on fuel injectors, fuel filters, water separators and labor before and after implementing this Best Practice.

Eliminating unscheduled truck downtime results in an increase in truck utilization. The benefits can be realized through either an O&O Cost savings or production gains. Note, only one of the two can be realized at any given time, not both.

For the example above, in years 1 and 2 production is held constant and therefore, the benefits result in lower O&O Costs. In other words, utilization increases, operating hours increase and O&O Cost per hour decreases.

Conversely, in years 3 through 5 production is increased and the benefits result in production gains. Now, since utilization increases, operating hours increase which results in increased productivity.

This is just one example of how the benefits can be captured using the table above. In some cases, increased production would be the only area of interest for benefit and likewise with O&O Cost savings. Obviously, the NPV value will vary depending on the scenario.

5.0 Resources Required

- Dedicated personnel to monitor fuel cleanliness at each stage.
- Bulk Fuel Filter Coalescer (if not already present).
- Particle Counter.
- Replacement Parts (hoses, fast fill connectors, screen filters, caps, pumps, etc.).

6.0 Supporting Attachments / References

Fuel Filtration Value Analysis Tool - Fuel Filtration Value Analysis Tool_Template.xls

7.0 Related Best Practices

0806-2.10-1000	Managing Fluid Cleanliness
1106-2.10-1037	Bulk Fuel Storage & Handling
0806-2.10-1006	Bulk Fuel Filtration

8.0 Acknowledgements

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